Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #46 – Remote Team Time-Tracking & Project Approver

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow developers to log time against specific project tasks and prevent timesheet submissions exceeding 24 hours in a single calendar day.	High	Pass: A valid timesheet is submitted for manager approval. Fail: A timesheet containing more than 24 hours in one day is accepted.	Prevents incorrect time entries and keeps timesheet records realistic.
FR-002	Functional	The system shall allow developers to submit their completed timesheets to the engineering manager for approval.	High	Pass: After submission, the timesheet appears in the manager's pending approval list. Fail: The submitted timesheet does not appear for the manager.	Provides a clear way to send recorded hours for approval.
FR-003	Functional	The system shall allow engineering managers to review submitted timesheets and either approve or reject them.	High	Pass: The manager can select Approve or Reject and the status is updated. Fail: The manager cannot change the timesheet status.	Supports the required managerial approval workflow.
FR-004	Functional	The system shall track the project budget burn rate based on the hours recorded against project tasks.	High	Pass: The project dashboard shows updated budget usage after new hours are recorded. Fail: The budget information remains unchanged after valid time is logged.	Helps managers see how quickly the project budget is being used.
FR-005	Functional	The system shall allow developers and engineering managers to view the current status of submitted timesheets.	Medium	Pass: The user can see statuses such as Pending, Approved, or Rejected. Fail: The current status is not displayed.	Keeps both developers and managers informed about the approval progress.
NFR-001	Nonfunctional – Performance & Security	The system shall update timesheet approval status transitions and project budget burn charts across dashboards within 500 ms.	High	Pass: Testing under simulated peak load shows dashboard updates within 500 ms. Fail: Updates take longer than 500 ms.	Provides fast access to current approval and budget information.
NFR-002	Nonfunctional – Security	The system shall restrict timesheet approval functions to authorized engineering managers.	High	Pass: Only users with manager permissions can approve or reject timesheets. Fail: A developer or unauthorized user can approve or reject a timesheet.	Prevents unauthorized changes to timesheet approvals.

UML Use-Case Diagram

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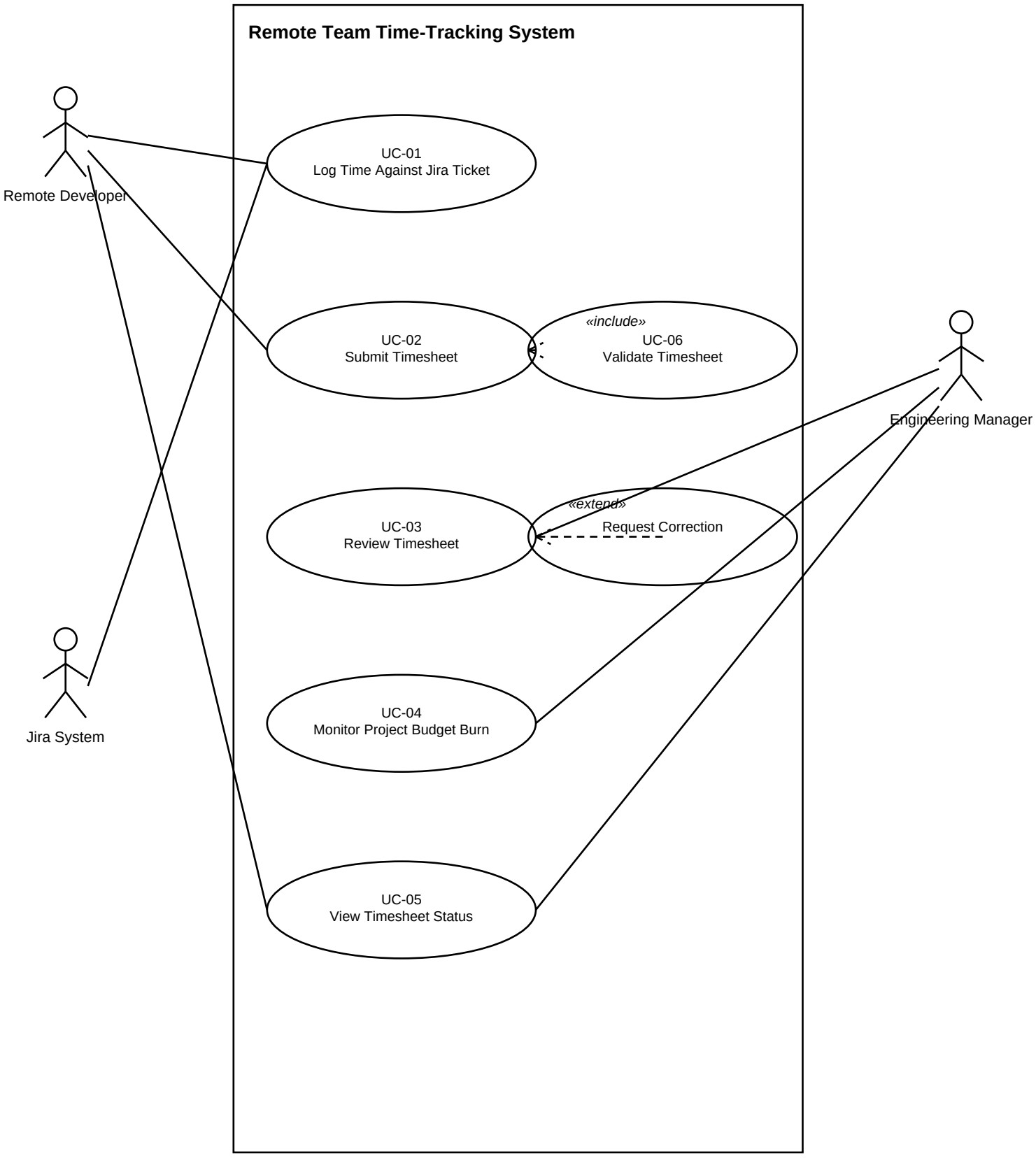

Solid line = association Dashed arrow = «include» / «extend»

Lab 1: Use-Case Flow & Supporting System Logic

Problem Statement #46 – Remote Team Time-Tracking & Project Approver

1. Use Case: Submit Timesheet

Primary Actor: Remote Developer
Secondary Actor: Engineering Manager
Supporting System: Jira System

2. Preconditions

- The developer is logged into the time-tracking system.
- The developer has recorded working hours against project tasks.
- The required task and project information is available.
- The timesheet is ready for submission.

3. Postconditions

- On success, the timesheet is stored and its status becomes Pending Approval.
- The Engineering Manager can view the submitted timesheet.
- On failure, the timesheet is not submitted and remains editable until the problem is corrected.

4. Main Success Scenario

1. The developer opens the Timesheet section.
2. The system displays the recorded time entries.
3. The developer checks or edits the entries.
4. The developer selects Submit Timesheet.
5. The system validates the timesheet data.
6. The system checks that no calendar day contains more than 24 hours.
7. The system accepts the submission.
8. The system stores the submission and changes its status to Pending Approval.
9. The Engineering Manager can see the submitted timesheet for review.
10. The use case ends successfully.

5. Alternate / Failure Flows

A. Daily hours exceed 24 hours: The system detects the invalid total, rejects the submission, shows an error message, keeps the timesheet editable, and asks the developer to correct the hours. After correction, the developer can submit again.

B. Required information is missing: The system identifies the missing field or entry, shows what needs to be completed, prevents submission, and allows the developer to correct the timesheet.

C. Manager rejects the timesheet: The manager reviews the submitted timesheet, selects Reject, and the system changes the status to Rejected. The developer can view the status, correct the entries if needed, and resubmit.

D. Unauthorized user tries to approve: The system checks the user's role. If the user is not an authorized Engineering Manager, the approval operation is denied and the timesheet status is not changed.

E. Jira task cannot be found or is unavailable: The system should not create a valid time entry against an unavailable task. It should show an error and allow the developer to select a valid task. This is a reasonable design assumption, not a condition explicitly stated in the problem statement.

F. System/update failure: If the submission cannot be stored successfully, the system should not show the timesheet as successfully submitted. The user should be informed that submission failed and should be allowed to retry.

6. What the System Validates Before Submission

- Developer identity and login.
- Project/task information for each time entry.
- Date and number of hours entered.
- Daily total must not exceed 24 hours.
- Required information must be present.
- User must have permission to submit the timesheet.

7. Manager Review Flow

1. The Engineering Manager opens the pending approval list.

2. The system displays submitted timesheets.
3. The manager selects a timesheet.
4. The system displays its entries and status.
5. The manager chooses Approve or Reject.
6. If approved, the status becomes Approved.
7. If rejected, the status becomes Rejected and the developer can correct and resubmit.

8. Simple Data / Database Schema

The Lab handout does not require a database schema or SQL commands. The following is included as supporting design knowledge in case the teacher asks what data the system would need.
USER : user_id (PK), name, role
PROJECT : project_id (PK), project_name, budget
JIRA_TASK : task_id (PK), project_id (FK), task_name
TIME_ENTRY : entry_id (PK), user_id (FK), task_id (FK), date, hours
TIMESHEET : timesheet_id (PK), user_id (FK), status, submitted_date
APPROVAL : approval_id (PK), timesheet_id (FK), manager_id (FK), status, approval_date

9. Main Relationships

• USER 1:M TIME_ENTRY
• JIRA_TASK 1:M TIME_ENTRY
• PROJECT 1:M JIRA_TASK
• USER 1:M TIMESHEET
• TIMESHEET 1:M TIME_ENTRY
• TIMESHEET 1:M APPROVAL
• Engineering Manager (USER) 1:M APPROVAL

10. Example Database Commands

These commands are only illustrative supporting knowledge; SQL/database commands are not listed as Lab 1 deliverables.
CREATE TABLE USER (user_id INT PRIMARY KEY, name VARCHAR(100), role VARCHAR(30));
CREATE TABLE PROJECT (project_id INT PRIMARY KEY, project_name VARCHAR(100), budget DECIMAL(12,2));
CREATE TABLE JIRA_TASK (task_id INT PRIMARY KEY, project_id INT, task_name VARCHAR(200));
CREATE TABLE TIME_ENTRY (entry_id INT PRIMARY KEY, user_id INT, task_id INT, date DATE, hours DECIMAL(4,2));
CREATE TABLE TIMESHEET (timesheet_id INT PRIMARY KEY, user_id INT, status VARCHAR(30), submitted_date DATE);
CREATE TABLE APPROVAL (approval_id INT PRIMARY KEY, timesheet_id INT, manager_id INT, status VARCHAR(30), approval_date DATE);

11. Important Points to Explain if Asked

• Why 24 hours? The problem statement explicitly says submissions exceeding 24 hours in one calendar day must be prevented.
• Why Pending Approval? A submitted timesheet has to go through the Engineering Manager's approval workflow.
• Why Reject? Managerial review naturally needs an approval/rejection outcome, and the requirement says the manager can approve or reject submitted timesheets.
• Why role checking? Approval should be restricted to an authorized Engineering Manager.
• Why include validation? Submit Timesheet always needs validation before a submission is accepted.
• Why extend Request Correction? Correction is conditional; it is needed only when the manager finds a problem.

12. Scope Note

The official Lab 1 deliverables are the requirements table, UML use-case diagram, and one-page use-case flow. The handout does not require database design or SQL. The extra failure cases and schema above are included only so the student can answer follow-up questions during discussion.